Miniaturized Mobile Robot for Educational Applications in Logistics

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***Abstract*—The internal movement of materials in industrial environments demands solutions that increase efficiency and reduce human errors. This work presents the development of a miniaturized prototype of a robotic forklift, built with Raspberry Pi, ultrasonic sensors, and LiDAR, designed for navigation and environmental perception according to the principles of Autonomous Vehicles. The system architecture is based on ROS 2, enabling the integration of sensors and the implementation of functionalities related to environment mapping and motion planning. Although the lifting mechanism is present, the complete capture of pallets remains as future work. The prototype demonstrates technical feasibility for educational and research purposes, contributing to Educational Robotics applied to logistics.**

***Index Terms*—Education Robotics, Autonomous Vehicle Navigation, Motion and Path Planning.**

I. Introduction

The internal movement of materials in industrial environments, commonly carried out by human operators using electric forklifts, presents challenges related to efficiency, safety, and operational costs. With the advances of Industry 4.0, it becomes possible to apply accessible technologies to automate these processes, promoting greater productivity and reducing human errors [1].

Digitalization and the use of cyber-physical systems, central pillars of Industry 4.0, enable real-time monitoring, distributed decision-making, and more flexible and efficient operations [2]. In addition, the adoption of these technologies contributes to the sustainable performance of organizations, reducing waste and increasing the reliability of processes [3]. Collaborative robotics also plays a relevant role in this context by increasing safety and precision in repetitive and risky tasks [4] [5].

In this context, this work presents the development of a miniaturized mobile robot designed as an educational resource to demonstrate concepts of robotics applied to logistics. The prototype integrates LiDAR and ultrasonic sensors with a Raspberry Pi computational unit, enabling the execution of movement tests, environmental perception, and initial operation in a physical mock-up representative of a logistics environment.

This article is organized as follows. Section II presents the theoretical foundations that support the development of the prototype. Section III describes the construction methodology, the integration of the components, and the use of ROS 2. Section IV presents the tests and validation of the developed system. Section V presents the cost estimation of the prototype. Finally, Section VI presents the final considerations and the future directions of the work.

II. Theoretical Background

This section presents the main concepts and technologies that support the development of the autonomous forklift prototype. Initially, the principles of Industry 4.0 and its role in the digital transformation of production processes are addressed. The characteristics and differences between Automated Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs) are also explored. In addition, the Robot Operating System (ROS) and SLAM (Simultaneous Localization and Mapping) techniques are discussed, which are fundamental for sensory integration, map construction, and autonomous navigation in mobile robots. These foundations provide the theoretical support necessary for understanding the proposal developed in this work.

A. Industry 4.0

Industry 4.0, also known as the Fourth Industrial Revolution, is characterized by advanced integration between physical and digital systems through technologies such as the Internet of Things (IoT), artificial intelligence (AI), big data, automation, autonomous robots, and cyber-physical systems [2]. These technologies provide greater interconnectivity, enabling real-time communication between machines, devices, and humans, leading to more flexible, efficient, and autonomous production [1].

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Real-time monitoring, decentralized decision-making, and dynamic adaptation to changes in production are fundamental characteristics of these advanced industrial environments, enabling highly personalized production that is responsive to market demands [6].

In addition to productive efficiency, Industry 4.0 brings significant benefits to supply chain and logistics management. Digital technologies applied to logistics, such as warehouse automation, robotics, autonomous vehicles, and intelligent monitoring, have improved operational efficiency and reduced costs, while also enhancing the ability to respond to consumer demands [1].

However, the implementation of these technologies is not free from challenges, especially in emerging markets such as Brazil, where structural and technological barriers may hinder broad and effective adoption. Even so, Industry 4.0 has great potential to significantly transform productivity and economic growth, also standing out for its positive impact on sustainability due to the more efficient use of resources and the reduction of waste in the production chain [1].

Thus, the proposed prototype applies the foundations of Industry 4.0 through the integration of sensors, embedded processing, and motion control, establishing an experimental basis for studies in logistics automation.

B. Automated Guided Vehicles and Autonomous Mobile Robots

Automated Guided Vehicles (AGVs) are autonomous systems widely used for the internal transport of materials, with navigation based on magnetic tracks, floor cables, or visual markers such as ArUco codes [7].

Due to the limitations of AGVs in dynamic environments, Autonomous Mobile Robots (AMRs) have emerged, using sensors such as LiDAR, RGB-D cameras, and SLAM algorithms to perform adaptive and safe navigation [8] [9]. These robots make decisions in real time and autonomously recalculate routes, which makes them ideal for variable industrial *layouts* and dynamic intralogistics applications, in which flexibility, route adaptation, and integration with the internal flow of materials are fundamental requirements [10] [11].

Recent studies also explore ROS-based navigation architectures for mobile robots, enabling the integration of sensors, motion control, and trajectory planning in autonomous systems [12]. Routing and scheduling techniques have also contributed to greater logistics efficiency [13].

In this context, the proposed prototype adopts AMR principles by integrating perception sensors and adaptive navigation strategies at a reduced scale, with an educational focus applied to logistics.

C. ROS and SLAM Mapping in Autonomous Mobile Robots

The Robot Operating System has become one of the main development architectures for mobile robots, as it provides a modular *middleware* based on *nodes*, *topics*, and *services*, enabling integration between sensors, actuators, and perception algorithms [14]. The ecosystem offers essential tools such as RViz, TF, and SLAM packages, which facilitate data acquisition, map construction, and autonomous navigation. In addition, embedded modular architectures based on Raspberry Pi and ROS 2 have also been used in the development of experimental robotic platforms [15]. Applications with visual SLAM on physical platforms also reinforce the flexibility of this ecosystem in mobile robotics [16].

Within this architecture, SLAM plays an important role by allowing the robot to localize itself while building a map of the environment, enabling navigation without external infrastructure. More recent studies show that current approaches incorporate more robust perception, integration between different sensors, and hybrid mapping strategies, making SLAM a central technology for autonomous mobile robots [14] [17] [18].

Another relevant approach consists of integrating LiDAR data with RGB cameras to improve obstacle detection. In addition, sensor fusion, such as LiDAR and ultrasonic sensors, has been used to increase the robustness of navigation in mobile robots [19]. Computer vision complements LiDAR by identifying elements that are not captured by the laser beam, such as very low objects or surfaces with low reflectivity. This sensory fusion makes navigation more robust and enables smoother and safer trajectories [20].

Thus, the adoption of ROS and the SLAM-based architecture in the prototype developed in this work enables the creation of a didactic platform capable of reproducing, at a reduced scale, the technological principles present in real industrial systems.

III. METHODOLOGY

This section describes the methodology adopted for the development of the miniaturized autonomous forklift prototype. The stages of design, modeling, assembly, and integration of the mechanical, electronic, and sensory components that make up the system are presented. In addition, the functional validation procedures carried out in this initial phase are detailed, aimed at verifying the correct operation of the mechanical structure, motors, sensors, and the embedded architecture.

A. Assembly of the Miniaturized Prototype

The construction of the miniaturized prototype initially involved the development of the three-dimensional technical design of the mini forklift chassis in a CAD (Computer-Aided Design) environment, followed by its fabrication through 3D printing using PLA filament. The project was developed considering the specificities of additive manufacturing, ensuring dimensional precision, structural resistance, and full compatibility with the electronic components that would be integrated into the system. In this way, the printed model presented adequate fittings, optimized internal organization, and full functionality for the assembly and operation of the mechatronic system.

The developed model uses four motorized wheels, each driven by an independent DC motor. This choice was adopted to ensure greater control over the robot's traction and to allow

more precise movements during displacement. By activating each wheel separately, the prototype gains stability and maintains direction more safely, which is an important requirement in reduced environments or with narrow trajectories. Steering is achieved through the combination of the rotation direction of the wheels on the left and right sides, following the principle of differential traction. In this way, the robot can perform natural turns and even rotate around its own axis, which facilitates its maneuvers and makes its movement smoother.

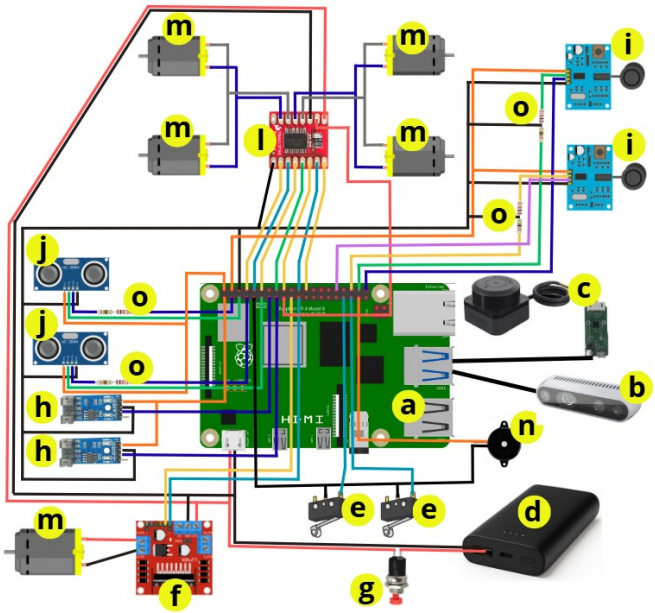


Figure 1: Robot Hardware and Electronic Architecture.

As shown in Figure 1, the components used in the prototype are presented below, along with a brief description of each element:

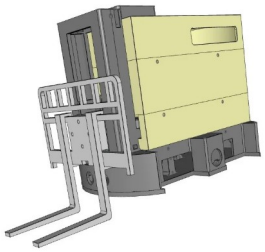
- (a) Raspberry Pi 4: It functions as the prototype's main processing unit, responsible for central system control, sensor data processing, decision-making, and communication among the modules. The device includes 8 GB of RAM and a 64 GB SD card used to store the operating system, ROS 2 packages, and telemetry logs.
- (b) Intel RealSense D435i: The Intel RealSense D435i Depth Camera is an advanced perception sensor widely used in robotics, combining a stereo depth camera system, an RGB camera, and an inertial measurement unit (IMU). This integration allows the capture of spatial and motion information in real time with high precision. In the prototype, the D435i will be used to identify the pallet and determine its relative position to the fork, providing a detailed depth map that enables more accurate alignment during the approach and the future execution of the pickup operation.
- (c) RPLiDAR C1 (360°) LiDAR Sensor: It performs real-time 2D scanning of the environment and is essential for map construction (SLAM) and for improving the precision of autonomous navigation.
- (d) 10,000 mAh (15W) Power Bank: Provides portable power to the system, ensuring sufficient autonomy for continuous operation during testing, with the capacity to supply energy to both sensors and motors.
- (e) Limit Switch Sensor: It will be used in the fork-lifting mechanism to detect the mechanical end of travel. This sensor identifies when the fork reaches the upper or lower movement limit, preventing motor overload and ensuring precise stopping at the end of its stroke.
- (f) L298N H-Bridge Driver: This module is used to control the rotation direction of the DC motors that operate the fork-lifting mechanism. It allows polarity inversion applied to the motors, enabling forward and reverse movement independently, as well as speed control through pulse-width modulation (PWM).
- (g) Power Button: This component functions as the main interface for turning the system on and off, allowing the operator to quickly initialize all electronic modules of the robot or safely interrupt its operation.
- (h) Optical Encoder Sensor and 20-Tooth Encoder Wheel: The system uses a set composed of a slot-type optical encoder sensor and a 20-tooth encoder wheel attached to the wheel shaft. The sensor detects the passage of the wheel's segments through optical interruption, generating digital pulses proportional to the shaft's rotation. This setup provides odometry, enabling the estimation of the distance traveled by the robot and contributing to the implementation of navigation and localization algorithms in ROS 2.
- (i) AJ-SR04 Ultrasonic Sensors: These sensors were installed on the sides of the prototype to expand the robot's perception field. Each module is responsible for detecting obstacles positioned alongside the vehicle.
- (j) HC-SR04 Ultrasonic Sensors: These sensors were installed on the front and rear of the prototype, significantly enhancing obstacle detection along the robot's main movement axis. This configuration allows the system to identify objects ahead during forward motion and monitor potential obstacles behind it, especially during reverse maneuvers.
- (l) TB6612FNG Motor Driver: This driver will be used to control the rotation direction and speed of the DC motors that make up the traction system. Since the entire assembly is powered at 5 V, the TB6612FNG offers excellent energy efficiency, with significantly lower losses compared to traditional drivers, ensuring better performance and reduced heat dissipation in the system.
- (m) DC Motors (3–6 V): A total of five DC motors will be used. Four independent motors will drive the wheels, enabling the prototype to move in all directions, with differential control of speed and rotation direction for executing turns and maneuvers. An additional motor will be dedicated to operating the fork, enabling the lifting and lowering of the simulated load.
- (n) Buzzer Sensor: This component is responsible for the system's audible alerts, emitting sound signals in

specific situations. It will be activated when the prototype is performing reverse maneuvers or when operating in mapped navigation mode and detecting an obstacle ahead, serving as an additional safety alert.

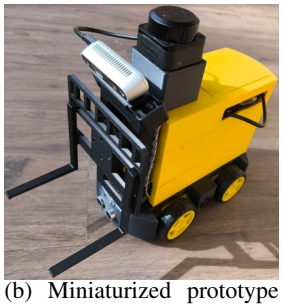
- (o) Resistor: 10 kΩ and 4.7 kΩ resistors were used to form the voltage divider required to reduce the AJ-SR04 ECHO signal to a safe level for the Raspberry Pi 4 GPIO pins.

To complement the description of the components and the proposed architecture, Figure 2 presents the mobile robot developed in this work.

Figure 2: Design and miniaturized prototype of the mobile robot.



(a) Robot design.



(b) Miniaturized prototype of the mobile robot.

The developed prototype has a total mass of 1.352 kg, a length of 24 cm, and a height of 22 cm. The width of the upper structure is 11 cm, excluding the wheels, while the total width of the assembly, considering the outer edges of the wheels, is 16.6 cm.

B. Software Development and Integration with ROS 2

The control and navigation system of the prototype was developed in Python, a language widely used in robotic applications due to its efficiency, portability, and native integration with ROS 2 (Robot Operating System). The choice of this architecture aims to ensure modularity, real-time communication, and ease of system expansion.

ROS 2, running on the Raspberry Pi 4 with the Ubuntu Server 22.04 operating system, acts as the communication core between all robot modules. It enables the distributed management of sensory and control information through the exchange of messages between nodes and topics. In the project, the LiDAR sensor was integrated into the ROS 2 environment to provide the data required for the environment mapping process and the initial stages of SLAM. Figure 3 presents the visualization of the initial SLAM process, highlighting the LiDAR readings used for the incremental generation of the map.

In the current state of development, the LiDAR sensor is fully integrated with ROS 2 and already provides continuous readings published on the /scan topic. The acquisition of these data was validated using RViz2, the standard tool for sensory visualization and debugging of spatial transformations (TF2).

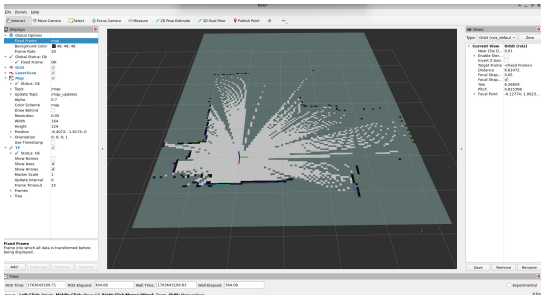


Figure 3: It illustrates the initial SLAM process visualized in RViz, showing the robot's first positioning estimates and the LiDAR readings used to begin the incremental construction of the map.

Considering that autonomous navigation has already been implemented at this stage, the validation of the readings in RViz2 allowed the correct acquisition and processing of sensor data to be confirmed, enabling the execution of the initial stages of SLAM. From these readings, it was possible to generate a 2D map of the environment and estimate the robot's location during the mapping process.

C. Energy Consumption of the System

The energy consumption of the prototype was estimated by considering all the components used. The four DC motors responsible for traction consume, on average, 1.0 A at 5 V, while the motor dedicated to the fork adds approximately 0.30 A. The Raspberry Pi 4 computational unit has an average consumption of about 1.0 A, and the perception sensors RPLiDAR C1 and Intel RealSense D435i add approximately 0.50 A and 0.20 A, respectively. The ultrasonic sensors and electronic drivers contribute an additional 0.20 A.

Thus, the system's average total consumption is approximately 3.0 A during operation. Considering the use of a 10,000 mAh power bank, the robot's estimated autonomy ranges from 2.5 to 3.0 hours, depending on the load applied to the motors. This consumption demonstrates that the prototype maintains a good balance between performance and energy efficiency, making it suitable for scale-model testing and educational applications.

IV. Testing and Validation

To validate the developed system, tests were carried out at different stages of the project, ranging from the assembly of the prototype to navigation experiments in the mapped environment.

A demonstration video of the proposed system is available at: `https://youtu.be/EFHZ58ks7_c`.

Initially, assembly tests of the prototype were conducted, verifying the correct positioning of the mechanical and electronic components, as well as the structural stability of the system. Subsequently, individual tests of the main devices connected to the Raspberry Pi 4 were performed, allowing validation of the isolated operation of sensors, motors, and control modules before full integration into the robot.

After the integration of the components, initial movement tests were carried out to adjust the motor control and calibrate the robot's odometry. The displacement of the platform was estimated from the wheel encoders, while the direction of movement was determined by the signals from the motor drive circuit, using the TB6612FNG driver. These tests made it possible to adjust the parameters of the differential kinematic model, improving the consistency between the robot's real movement and the estimation of its pose.

Obstacle detection tests were also carried out using ultrasonic sensors positioned at the front, rear, and sides of the robot, with the objective of validating the distance measurement capability and supporting basic collision avoidance strategies.

In the next stage, the software modules responsible for environment mapping were implemented and tested using ROS 2 and the RViz tool. From the integration between the sensors and the odometry system, it was possible to generate a two-dimensional map of the environment. With the obtained map, localization and navigation tests were performed using the *2D Pose Estimate* and *Nav2 Goal* features, allowing the robot to move autonomously to positions defined on the map. Figure 4 presents the environment used in the experiments and the visualization of the map in RViz during the robot navigation tests.

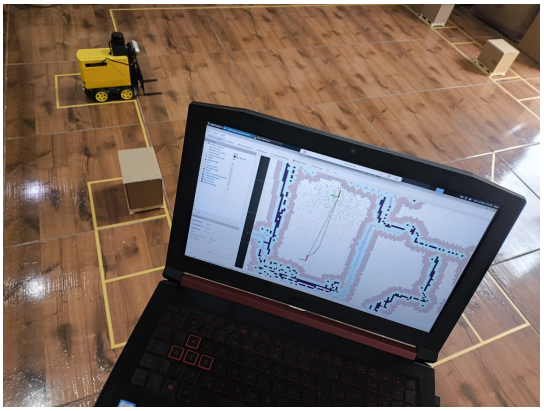
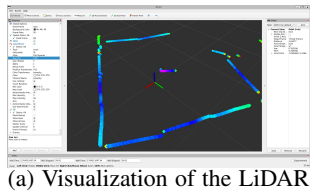


Figure 4: Test environment and map loaded in RViz for conducting the robot navigation tests.

In addition, energy autonomy tests were conducted in which the robot operated continuously in obstacle avoidance mode. Under this constant operation scenario, the prototype remained active for approximately 2.5 to 3.0 hours until the battery was fully discharged, indicating that the energy system is adequate for experiments in a mock-up environment and for educational applications.

Figure 5 presents the initial moment of detection and execution of the obstacle avoidance maneuver during the robot's navigation. When identifying an object in front of it through the LiDAR readings, the system automatically initiates a change of direction to avoid the collision, subsequently resuming its trajectory in the environment. The results demonstrate the feasibility of the integration between sensors, actuators, and the navigation system.

Figure 5: Execution of the obstacle avoidance maneuver by the mobile robot after detecting an object ahead.



(a) Visualization of the LiDAR data in RViz, showing the initial detection of the obstacle and the identification of the free area for maneuvering.

(b) Beginning of the avoidance maneuver, with the robot initiating the change of direction after recognizing the object ahead.

Although the initial results demonstrate the feasibility of the integration between the sensors, actuators, and navigation system, the process still requires improvements, especially with regard to odometry accuracy and navigation stability.

V. Cost Estimation

Table 1 presents the details of the main components used in the construction of the miniaturized autonomous forklift prototype, as well as their respective acquisition costs, including sensors, actuators, the processing unit, and additive manufacturing materials.

Table I: Costs of the components used in the miniaturized prototype (values in USD).

Component	**Cost (USD)**
Intel RealSense D435i	694.75
Raspberry Pi 4 (8 GB)	131.36
Power Bank 10,000 mAh (15W)	28.73
DC Motors 3–6 V (5 units)	35.92
L298N H-Bridge Driver	4.65
TB6612FNG Driver	12.31
Optical Encoder + 20-Tooth Disk (2 units)	8.37
HC-SR04 (2 units)	7.59
AJ-SR04 (2 units)	17.14
RPLiDAR C1 (360°)	160.09
Limit Switch	3.80
Buzzer	2.26
Cables and Connectors	15.19
Filament + 3D Printing	53.88
Total	**1176.04**

The total value presented provides an estimate of the cost of the main components used in the construction of the prototype and highlights the possibility of developing experimental mobile robotics platforms using commercially available hardware, facilitating their replication in educational and research environments.

The costs are presented in US dollars (USD) to facilitate international comparison and reproducibility, since many robotics components are globally commercialized and referenced in this currency.

VI. Final Considerations

Logistics automation based on autonomous mobile robots is a central element of Industry 4.0, promoting greater efficiency, safety, and operational autonomy. The miniaturized prototype developed in this work demonstrated the ability to integrate sensors, embedded electronics, and ROS 2 in a functional manner, reproducing, on a reduced scale, fundamental principles of navigation, environment perception, and lifting operation. The initial implementation of SLAM established the foundation for mapping and for future structured navigation strategies.

Although the project is still in a phase of continuous validation, the system proved functional for data collection, initial map generation, and preliminary obstacle-detection tests, demonstrating potential for academic and industrial applications. Obstacle avoidance was successfully achieved after integrating the motors, sensors, and embedded system, when initial movement tests were conducted using a simple avoidance algorithm. In these experiments, the robot moved through the environment, detected barriers ahead, and executed basic avoidance maneuvers, resuming its trajectory after preventing a collision, which demonstrated the effectiveness of the embedded architecture. These results consolidate a robust starting point for the next stages of development.

As a continuation of this work, enhancements to the SLAM system are planned, enabling the generation of more accurate maps as the robot moves through the environment and allowing the transition to predefined routes and navigation between specific points. The next major advancement involves developing the pallet-handling process through computer vision techniques and sensor fusion. The Intel RealSense D435i camera will be used to identify the pallet, estimate depth, and guide the alignment of the forks, while additional sensors such as the front HC-SR04 will provide complementary measurements to ensure a safe approach.

These improvements are expected to enhance the overall performance of the prototype and bring it closer to the functionalities found in real industrial logistics automation systems.

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